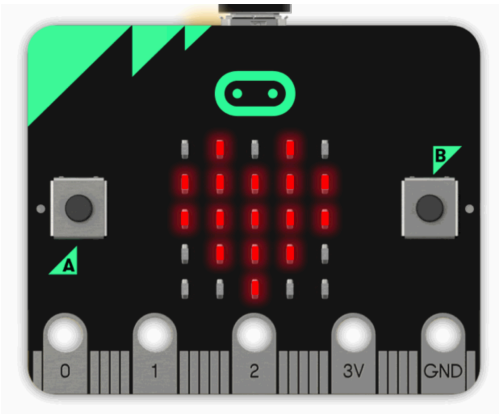
Lab 3: Interrupts



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Contents

1. Introduction 1
2. Tasks 2

ⓘ **Much to read, not as much to do. Read *before* the scheduled lab session!**

1. Introduction

This lab begins with a pair of programs that output a list of primes on the serial port. One of the programs uses polling to wait for the serial port to be ready before transmitting each character; the other buffers the characters waiting to be output, and uses interrupts to send each character when the port is ready.

The lab3-primes directory of the lab materials contains the following files:

File Name	Description
`Makefile`	Build script
`primes-poll.c`	Main program that uses polling
`primes-intr.c`	Main program that uses interrupts
`hardware.h`	Header file with layout of I/O registers
`lib.c, lib.h`	Library with implementation of printf
`startup.c`	Startup code
`nRF51822.ld`	Linker script
`lab3.geany`	Geany project file

Typing `make` as usual (or selecting Build>Make in Geany) will build two version of the primes program: in `primes-poll.hex` is the version that uses polling, and in `primes-intr.hex` is the interrupt-driven version. The implementation of the function `serial_putc` that does the work of `printf` is different in the two programs, and the interrupt-driven program has an additional function with the special name `uart_handler` that the hardware calls when a UART interrupt is triggered.

One of the LEDs on the micro:bit is turned on while the program is running and printing the first 500 primes, and turned off at the end. You can time the program with a watch, or wire the board up to an oscilloscope or logic analyser to get a more accurate timing.

2. Tasks

- Modify `primes-poll.c` so that transmission of each character completes before `serial_putc` returns, rather than before transmitting the character on the next call. Does this have a measurable effect on the running time?
- How small can you make the transmit buffer and still have the interrupt-driven version `primes-intr.c` work? Does a very small buffer adversely affect the running time?
- Add code to monitor the maximum number of characters stored in the transmit buffer, and print it at the end. Try increasing the buffer size to larger powers of two – you should be able to use values up to 8192 – and see if the whole buffer is ever filled.
- Add code to turn on an LED when the program is searching for the next prime, and turn it off when it is printing a prime it has found. Use a scope or logic analyser on the LED and the serial line to visualise the overlap between thinking and printing.
- What happens to the running time of both programs if we modify them to print not the first 500 primes, but the first 500 primes that are more than 1 000 000 or 10 000 000?
- If the calls to `intr_disable` and `intr_enable` in `serial_putc` are removed, does the program continue to work? Can you persuade it to go wrong? What if the critical section is reduced to cover only the `else` part of the conditional `if (txidle)`?
- When an interrupt occurs, the register values are saved on the stack. This ought not to affect the functioning of properly written code that has been correctly translated: but can you write some sneaky code to detect that memory just beyond the top of the stack is changing in an unpredictable way? Hint: the loop in `serial_putc` ought to experience some interrupts.

More demanding:

- Find out from Chapter 21 of the hardware reference manual for the nRF51822 how to configure the random number generator. Write a driver based on the hints given in Problem Sheet 3, and write a program that generates and prints a histogram showing the distribution of random values. Note: symbolic constants for the device addresses of the RNG are in the latest revision of the source file `hardware.h`.
- A bizarre challenge: find out if serial transmission can be implemented by bit-banging. This will mean configuring the correct pin as a GPIO output, and using delay loops to generate the RS-232 waveform with the correct timing. Use of an oscilloscope or logic analyser will be essential to get the timing right.

TODO: Update to be explicit about requirementes for S and S+.